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Introduction

Microfluidics is the study of fluids in channels on the order of magnitude of micrometers. This scale has important properties, most notably being that fluid behaviour is governed primarily by viscous forces, rather than inertial effects seen in larger scale fluid dynamical systems, leading to relatively predictable laminar flow compared to turbulent flow more commonly seen in larger systems. The set of experiments detailed in this lab report explore basic fluid mechanics concepts by using a fluid with fluorescent beads for image tracking. We will analyze flow in straight channels, channels with a change in width, and curved channels. The key equations used in this lab include the Hagen-Poiseuille equation (1), which describes volumetric flow rate Q as

$$Q = \frac{S^2(\Delta p + \rho g \Delta z)}{8\pi\mu L} \tag{1}$$

where S is the channel cross sectional area, Δp is the pressure difference from the starting to end point, ρ is the fluid density, g is the gravitational acceleration constant, Δz is the height difference from the starting to end point, μ is the dynamic velocity, and L is the channel length.

We will also use Torricelli's law (2), which relates changes in height to exit velocity.

$$U = \sqrt{2g\Delta z} \tag{2}$$

Experimental procedures

We followed the procedures from the lab manual [1] to record all data. Beaded fluid flowed from a syringe through the channels on the microfluidic chip. We recorded data for the straight channel at 5 different heights, as well as at a gradual and sharp change in width channel, and a right angled and smooth angled bend. For finding changes in height, we used a ruler to measure the height of the bottom of the fluid reservoir to a fixed point for easier measurements and then measured the height difference from the fixed point to the chip to find the actual change in height. The data was analyzed using the internal metadata files to find exposure time as well as the pixel/mm ratio. The images were analyzed in Microsoft paint to count the length of a streak in pixels, which was then multiplied by the pixel/mm ratio to find distance. We then used equation (3) to find the velocity of the fluid.

$$v = \frac{d}{t} \tag{3}$$

Error analysis

The uncertainty in velocity was largely from the measurement of streak lengths. Streak lengths that were expected to be similar were observed to have a random (i.e. non-systemic) variance. Most streaks were within ± 2 pixels of each other, so that was the uncertainty used in length of streaks. The exposure uncertainty was taken to be $\pm$ 0.1 ms or 1 count of the least significant digit. The relative uncertainty in velocity was therefore propagated as follows:

$$\frac{\Delta v}{v} = \sqrt{\left(\frac{\Delta l}{d}\right)^2 + \left(\frac{\Delta t}{d}\right)^2} \tag{4}$$

For all data collected, this value was 0.1 mm/s to one significant figure.

For the nozzle height measurements, we made two different ruler measurements, and therefore the uncertainty in measurements would be half the smallest increment of measurement (0.5mm) twice, which would be 1mm.

The distance to the centreline had larger uncertainties, as there was uncertainty in determining the reference centreline and then the distance from a streak to the reference. After multiplying by the scale factor, we took this uncertainty as 0.01 mm.

In a zoomed in image of the channel walls, we observed that the largest imperfection size was about 23.7 μm. This would exert a small force on the fluid perpendicular to the direction of fluid flow. This would theoretically cause some turbulent flow near the walls, but this effect was hard to see due to the resolution of the images and therefore probably negligible. The velocity would have a slight decrease near the wall due to kinetic energy being dissipated as friction but would have a negligibly small effect on the mean velocity along the streamwise direction. Given that there was energy dissipation, there would be decrease in downstream pressure (assuming the other variables are held constant). There was also a bubble or crack at the entrance, which would have shrunk local channel width, and therefore increased pressure and velocity.

We made assumptions of incompressible Newtonian flow as well as constant density, which is a reasonable approximation but obviously untrue as the beads were of different material than water. As there was a low density of beads, there would have been slight shear effects, which would have decreased effective viscosity, but that effect should be small.

Overall, the combined effect of these factors was relatively minor and did not affect the results to a significant extent as they still matched the expected results within the uncertainty.

Results and discussion

Straight channels

The recorded flow in the straight channel demonstrated continuous laminar flow. The fluid velocity at the channel wall approached zero, which is consistent with the no slip boundary condition. There was no visual evidence of turbulent flow. From theory, expect the maximum velocity to be found at the centre of the channel, and the maximum velocity gradient to be found at the channel walls. Both of these results are due to the increased friction at the channel walls. We observe this behaviour in our images, which can also be seen in figure 1, the velocity profile of the fluid with uncertainties. We expect the curve of the velocity profile to be parabolic, which is the observed result in the measured profile.

Based on Bernoulli's equation, we can change the mean velocity in the straight channel by changing the height of the syringe (greater height difference leads to greater velocity), the fluid density (denser fluids would have lower velocity), and pressure difference (increased pressure difference leads to increased velocity). The only parameter we can change in this lab setup was the syringe height, and the results of that were shown in figure 2.

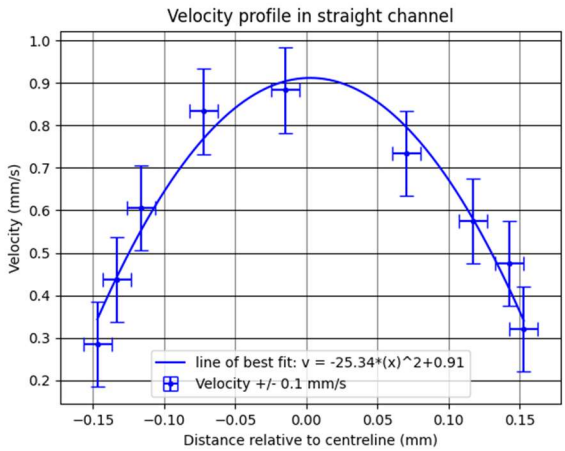

Figure 1 Velocity profile with uncertainties plotted showing a parabolic relationship between position and velocity. The uncertainty in position is ± 0.01 mm *and the uncertainty in velocity is* $\pm 0.1\ mm/s$

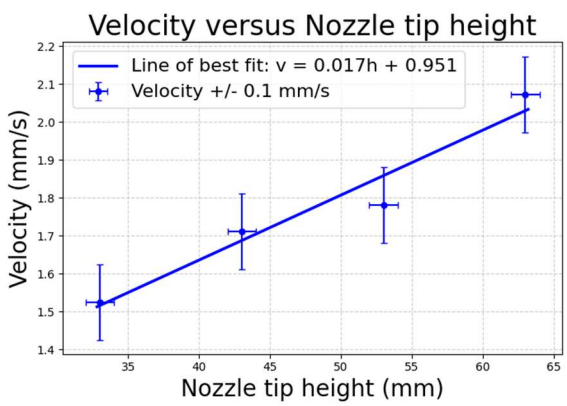

Figure 2 Flow velocity versus nozzle tip height. We observe a linear increasing relationship. The uncertainty in tip height is $\pm 1mm$ *and the uncertainty in velocity is* $\pm 0.1\ mm/s$

To find the relationship between syringe height and straight channel velocity, we used equation (1). By conservation of mass, the flow rate everywhere is constant at steady state. From the setup provided [1], we find the pressure difference between the top of the fluid in the syringe and the exit to be equal to the sum of viscous losses across all individual sections.

$$\mathrm{P}_1 - P_6 = Q \times \sum_i \left(\frac{8\pi\mu L_i}{S_i^2} - \rho g \Delta z_i \right) \quad (5)$$

Since the pressure at the top of the fluid and at the exit are both equal to the atmospheric pressure, the left-hand side of (5) is 0. Δz_i is the net difference in height between the top of the syringe and the exit, or $z_1 - z_6$. We can use the fact that in the straight channel, $Q = S_3 U_3$ (from [1]), and then plugging in and rearranging for U_3 we get

$$U_3 = \frac{\rho g (z_1 - z_6)}{S_3 \sum_i \frac{8\pi\mu L_i}{S_i^2}} \quad (6)$$

All the non-z terms in the right-hand side are constant for the given set up, so therefore the relationship is of linear proportionality, or $U_3 \propto (z_1 - z_6)$. This aligns with the observed linear increasing relationship from figure 2.

Channels of different size

The channel width before the change is $w_1 = 0.703$ mm. The channel width after the width change is $w_2 = 0.327$ mm. We assume that the channel thickness in the Z direction (tangent to chip surface) is constant. We are also assuming that the fluid behaves as an incompressible Newtonian fluid undergoing laminar flow, which is a pretty reasonable assumption for the centreline velocity. As well, we assume that there are no edge/entry

effects, and that the flow has reached a steady state.

The relationship between the flow rate (Q) and the average velocity (v) and cross-sectional area (A) is given in equation 7.

$$Q = v \times A \tag{7}$$

As the flow profile was determined to be parabolic, the average velocity is half the maximum, or centreline velocity. As we have a constant height and flow rate (assuming conservation of mass applies), we can rearrange equation 7 to find the ratio of centreline velocities from the ratio of widths.

$$\frac{v_1}{v_2} = \frac{w_2}{w_1} = 0.465 \tag{8}$$

The actual measured velocity ratio was determined as follows.

$$\frac{v_1}{v_2} = \frac{0.00116}{0.00183} = 0.633 \tag{9}$$

Equations 8 and 9 are not in agreement with each other. There is 50% error, which is too big to be accounted solely in measurement uncertainty. The most likely source of the discrepancy is that the streaks were not indicative of actual flow velocity. This is likely as in other images, we observed that streak length was not perfectly correlated with expected velocity (i.e. that streaks that are supposed to be longer (at points with narrower widths for example) were actually shorter than those at wider width values). This physically could have been due to local clumping of beads, or the camera not picking up the streaks properly (we did increase the gain to be able to view the streaks, which would have increased the signal to noise.

In the channel with gradual transitions (figure 3), the streaks were consistent and present throughout the width of the channel. The velocity increased gradually as the width narrowed It was harder to tell for the channel with an abrupt transition because there was a large bubble/crack right at the entrance, but looking at the unobstructed half, we can visibly see a velocity increase, and can also see that near the right angled corner there exists non laminar flow, where eddies are created that are not entirely going into the smaller channel, whereas in the gradual channel essentially all fluid went flowed into the smaller channel.

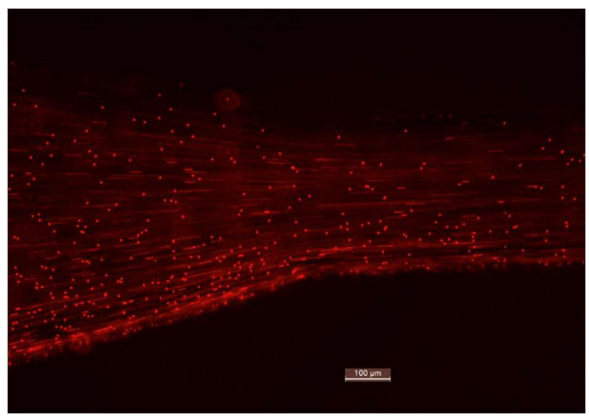

Figure 3 Channel with a gradual width transition. The flow remains smooth throughout.

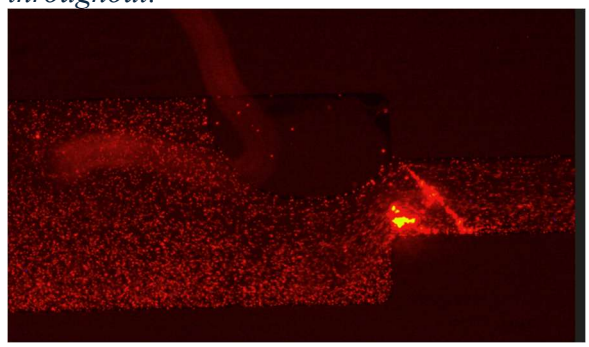

Figure 4 Channel with an abrupt width transition. There is turbulent flow at the right angled corner.

For the smooth transition one channel, there is almost entirely flow. The only place where the flow isn't super smooth is at the edges due to edge friction, but that is a minimal effect compared to the rest of the laminar flow.

For the abrupt transition channel seen in figure 4, the flow is harder to describe, especially with the larger crack/bubble removing one side of the abrupt transition, but there appears to be turbulent flow, particularly the flow that is forced up the corner. The flow that isn't flowing isn't technically turbulent but also not laminar, it is hard to classify due to the energy losses and associated low speed and therefore it is hard to see how it changes direction over time.

Flow transitions are common because we have different sized channels for different purposes (particularly from bulk storage and transportation into individual uses). Smooth transitions have more predictable flow so there would be less energy loss, resulting in an overall more efficient system.

Channels with Bends

From conservation of mass and equation (7), if the cross sectional area remains the same, then the velocity should remain the same, so the expected centreline velocity ratio before and after the bend would be 1. In the right angled bend channel seen in figure 5, the entry centreline velocity was 0.904 mm/s, and the exit velocity was measured as 0.984 mm/s. For the channel with an acute angled bend seen in figure 6, the entry speed was measured to be 0.267 mm/s, and the exit speed was measured as 0.354 mm/s

The ratio of entry to exit velocity in the channel with a right angled bend is therefore $\frac{0.904}{0.984} = 0.918$, and the velocity ratio in the channel with an acute angled bend is $\frac{0.267}{0.354} = 0.754$. Both of these results disagree with conservation of mass but agree with the theory that vortex interactions would increase the velocity [1]. As the uncertainty in velocity is $\pm 0.1 mm/s$, the values do agree with conservation of mass within the uncertainty, however as we are looking for a qualitative result, we observe that the streak length does increase at the channel exit.

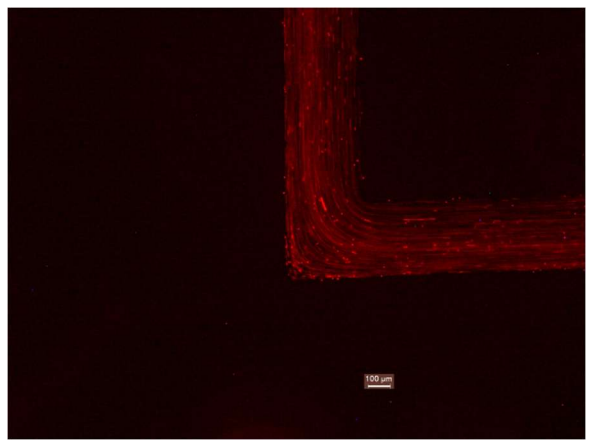

Figure 5 Flow in a channel with a right angled bend.

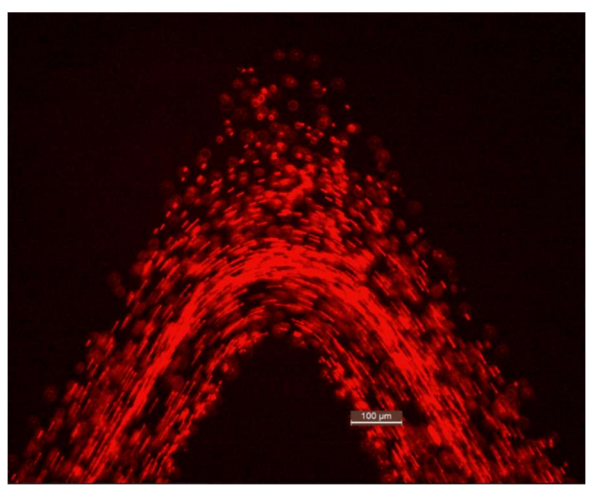

Figure 6 Flow in a channel with an acute bend.

The flow in the right angled bend channel creates a little bubble at the inside corner exit, the flow that passes over this bubble remains laminar. There is very little flow separation and no similar bubble in the acute bend channel. The concentration of particles does spread out a bit in the acute bend channel compared to the right angled bend channel due to the increase in cross sectional width at the corner apex of the acute bend channel.

There is very little observed turbulent flow In either channel, even near the corner. Although there is a small pocket of air observed at the inside corner exit of the right angled bend, which would indicate some turbulent flow, no vortices are observed. This is more likely due to limitations in imaging rather than evidence of pure laminar flow.

Discussing the corners of the right angled bend channel further, we see that the outer corner has a larger radius of curvature, which effectively increases the local "cross sectional area". Inertia in flow means that flow doesn't fully spread out to become Newtonian, so locally flow slows down due the increased effective cross section. The flow loses energy at the outside part of the corner, and the velocity of flow at this point is almost stationary (again, hard to tell with current analysis setup).

At the inside corner, there is a smaller radius of curvature. Thus, the effects are the opposite of those mentioned in the previous paragraph, so the flow that is not affected by wall friction becomes faster. The net effect of this is that the maximum velocity is not at the midpoint of the cross section, but rather closer to the inside corner. Additionally, the flow creates small bubble on the inside corner wall at the exit as it cannot fully stick to a 90 degree bend.

Conclusions

This series of experiments measured flow through different geometries of a microfluidic chip. We found that the flow profile was parabolic in a straight channel, which is in agreement with the Newtonian theory of flow. As well, we found that the flow velocity varied linearly with changes in nozzle height, which is in agreement with the Hagen-Poiseuille equation. We found that the flow velocity changed as the channel width changed, but the numerical results were not in agreement with the theory. Lastly, we found that the flow velocity in a channel with bends could be modelled more accurately by considering vortex interactions instead of conservation of mass. Our primary source of error was that the streaks in images did not perfectly correlate with flow velocity. We observed this qualitatively, which could have been for a variety of physical reasons, or due to imaging limitations. Our results agreed for the most part with the theory within uncertainty.

Bibliography

[1] P. Deng and Z. Lu, *Microfluidics Lab Manual, AER210,* University of Toronto, 2025.

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Appendix

If we use Bernoulli's equation, we make simplifying assumptions to reduce the equation to Torricelli's law (equation 2). As we assume gravitational acceleration is constant, we get the following relationship between mean channel velocity (U_3) and syringe height (z_1) (chip height z_6 is constant).

$$U_3 \propto \sqrt{z_1 - z_6} \tag{10}$$

This is in disagreement with the linear relationship of the Hagen–Poiseuille model. The two relationships differ because Bernoulli's equation neglects viscosity, and assumes all gravitational potential energy converts into kinetic energy. The Hagen–Poiseuille model accounts for energy loss in viscous dissipation, which is proportional to flow rate. The viscous dissipation increases the change in pressure, which increases the flow velocity. At the scale of this experiment (microfluidic), flow forces are governed by viscosity, so the linear relationship matches the experimental data. The Bernoulli relation be more applicable in a larger system, where viscous forces are not dominant.